



Ext 2

NORTH SYDNEY BOYS HIGH SCHOOL
2009
ASSESSMENT TASK 1

Mathematics
Extension 2

General Instructions

- Working time – 60 minutes
- Write on one side of the paper (with lines) in the booklet provided
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Barrett
☐ Mr Fletcher
☐ Mr Weiss

Student Number: _____

(To be used by the exam markers only.)

Question No	1	2	3	4	Total	Total
Mark	$\frac{\quad}{20}$	$\frac{\quad}{15}$	$\frac{\quad}{9}$	$\frac{\quad}{8}$	$\frac{\quad}{52}$	$\frac{\quad}{100}$

Question 1 (20 marks)

- (a) If $z = 1 + 2i$ and $w = 3 - i$, find in the form $a + ib$
- (i) $z \overline{w}$ 2
- (ii) $\frac{z}{w}$ 2
- (b) (i) Express $-1 + i$ in modulus-argument form 2
- (ii) **Hence** evaluate $(-1 + i)^{11}$, giving your answer in the form $a + ib$. 2
- (iii) What is the least positive integer for which $(-1 + i)^n$ is a positive integer? 1
- (c) Sketch the following loci on an Argand diagram:
- (i) $|z + 3 - 4i| = 5$ 2
- (ii) $|z| \leq |z - (2 + 2i)|$ and $-\pi \leq \text{Arg } z \leq \frac{\pi}{4}$ 3
- (d) (i) Show that $\sqrt{8 - 6i} = \pm(3 - i)$ 3
- (ii) Hence solve $6z^2 - 4iz + (i - 2) = 0$, giving answers in the form $a + ib$. 3

Question 2 (15 marks)

- (a) Consider the polynomial $P(z) = z^4 - 4z^3 + 6z^2 - 4z + 5$.
- (i) Show that $z + i$ is a factor of $P(z)$. 2
- (ii) Hence state why $z - i$ is also a factor of $P(z)$. 1
- (iii) Form the monic quadratic equation in z whose roots are $\pm i$. 1
- (iv) Hence factorise $P(z)$ fully over the complex numbers. 2
- (b) Given that the polynomial $P(z) = z^3 + 2z^2 - z + 1$ has zeros α, β and γ , find the polynomial function whose roots are
- (i) $\alpha - 1, \beta - 1$ and $\gamma - 1$ 2
- (ii) $(\alpha - 1)^2, (\beta - 1)^2$ and $(\gamma - 1)^2$ 2
- hence (iii) find the value of $\frac{1}{(\alpha - 1)^2} + \frac{1}{(\beta - 1)^2} + \frac{1}{(\gamma - 1)^2}$. 2
- (c) Using the polynomial in part (b), form the polynomial which has roots $\alpha\beta, \beta\gamma$ and $\alpha\gamma$ 3

Question 3 (9 marks)

- (a) When a polynomial is divided by $x - 2$ and $x - 3$, the respective remainders are 4 and 9. Find the remainder when the polynomial is divided by $x^2 - 5x + 6$. **3**
- (b) (i) If the polynomial $P(x)$ has a zero α of multiplicity n , prove that $P'(x)$ has the same zero with multiplicity $n - 1$. **3**
- (ii) If $ax^3 + bx^2 + d = 0$ (where $d \neq 0$) has a double root, show that **3**
 $27a^2d + 4b^3 = 0$.

Question 4 (8 marks)

- (a) (i) If $z = \cos \theta + i \sin \theta$, use de Moivre's theorem to show that **2**
 $z^n + z^{-n} = 2 \cos n\theta$.
- (ii) Hence solve $2z^4 + 3z^3 + 5z^2 + 3z + 2 = 0$. **3**
- (b) If $|z - \omega| = |z + \omega|$, find the difference between $\text{Arg } z$ and $\text{Arg } \omega$. **3**
Show all reasoning.

Q1) a) i/ $z\bar{w} = \frac{(1+2i)(3+i)}{1+7i}$

ii/ $\frac{z}{w} = \frac{1+2i}{3-i} \times \frac{3+i}{3+i}$
 $= \frac{1+7i}{10}$
 $= \frac{1}{10} + \frac{7i}{10}$

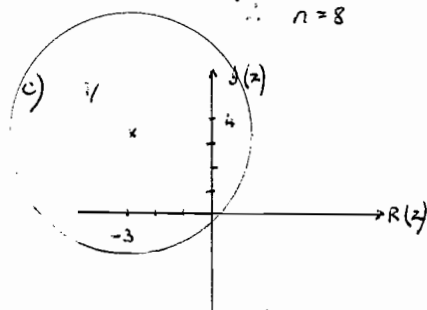
b) i/ $-1+i = \sqrt{2} \cos \frac{3\pi}{4}$

ii/ $(-1+i)^{11} = (\sqrt{2})^{11} \cos \frac{33\pi}{4}$
 $= 32\sqrt{2} \cos \frac{\pi}{4}$
 $= 32\sqrt{2} \left(\frac{1}{\sqrt{2}} + \frac{i}{\sqrt{2}} \right)$
 $= 32 + 32i$

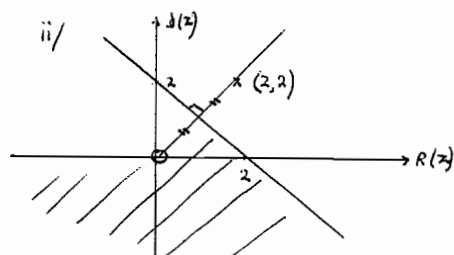
iii/ $n \cdot \frac{3\pi}{4} = 2k\pi$, k integer

$n = \frac{8k}{3}$

first integer n when $k=3$
 $\therefore n=8$



circle centre $(-3, 4)$ radius 5



• arg
 • mod
 • form } 1 off each error

d) i/ $\sqrt{8-6i} = a+ib$

$\therefore a^2 - b^2 = 8$
 $ab = -3$
 $a = \frac{-3}{b}$

$\therefore \frac{9}{b^2} - b^2 = 8$

$b^4 + 8b^2 - 9 = 0$

$(b^2+9)(b^2-1) = 0$

$\therefore b = \pm 1$ only

when $b=1$ $b=-1$
 $a=-3$ $a=3$

$\therefore \sqrt{8-6i} = \pm (3-i)$

ii/ $6z^2 - 4iz + (i-2) = 0$

$\therefore z = \frac{4i \pm \sqrt{16i^2 - 24(i-2)}}{12}$

$= \frac{4i \pm \sqrt{32-24i}}{12}$

$= \frac{4i \pm 2\sqrt{8-6i}}{12}$

$= \frac{2i \pm (3-i)}{6}$

$= \frac{6i-6}{6}$ or $\frac{i+3}{6}$

$= \frac{i-1}{2}$ or $\frac{i+3}{6}$

- region 1 $y < 2-x$
- region 2 $-\pi \leq \arg z \leq \frac{\pi}{4}$
- correct region shaded
- open circle

i off per error

Q(2a) i) $P(-i) = (-i)^4 - 4(-i)^3 + 6(-i)^2 - 4(-i) + 5$ (1)
 $= 1 - 4i - 6 + 4i + 5$
 $= 0$ (2)

ii) The coefficients of $P(z)$ are real. $\therefore$ complex roots come in conjugate pairs $\therefore z-i$ is a factor

iii) $z^2 + 1 = 0$ (1)

iv) $P(z) = (z^2 + 1)Q(z)$ (1)

$$\begin{array}{r} z^2 - 4z + 5 \\ z^2 + 1 \overline{) 2^4 - 4z^3 + 6z^2 - 4z + 5} \\ \underline{2^4 + z^2} \\ -4z^3 + 5z^2 - 4z \\ \underline{-4z^3 - 4z} \\ 5z^2 + 5 \\ \underline{5z^2 + 5} \\ 0 \end{array}$$
 (2)

$\therefore P(z) = (z^2 + 1)(z^2 - 4z + 5)$ (1) $z = \frac{4 \pm \sqrt{16 - 20}}{2}$

b) i) $P(z+i) = 0$ $(z - (2+i))(z - (2-i))$ (1) $= \frac{4 \pm 2i}{2} = 2 \pm i$

$\Rightarrow (z+i)^3 + 2(z+i)^2 - (z+i) + 1 = 0$

$z^3 + 3z^2 + 3z + 1 + 2z^2 + 4z + 2 - z - 1 + 1 = 0$ (2)

Q(2) ~~24~~ $z^3 + 5z^2 + 6z + 3$

ii) ~~P~~ $Q(\sqrt{z})$

$= (\sqrt{z})^3 + 5(\sqrt{z})^2 + 6\sqrt{z} + 3 = 0$

$= z\sqrt{z} + 5z + 6\sqrt{z} + 3 = 0$ (2)

$\sqrt{z}(z+6) = -5z-3$

$z(z+6)^2 = (5z+3)^2$

$z(z^2 + 12z + 36) = 25z^2 + 30z + 9$

$R(z) = \underline{z^3 - 13z^2 + 6z - 9 = 0}$

b) iii) $R\left(\frac{1}{z}\right) = 0$

$$= \left(\frac{1}{2}\right)^3 - 13\frac{1}{2^2} + \frac{20}{2} - 9 = 0$$

$$= \frac{1}{23} - \frac{13}{22} + \frac{20}{2} - 9 = 0$$

$$1 - 13z + 20z^2 - 9z^3 = 0$$

Or $2z^3 - 20z^2 + 13z + 1$

h

c)

$$P(x) = z^3 - (\alpha\beta + \beta\gamma + \alpha\gamma)z^2 + (\alpha\beta\gamma + \alpha^2\beta + \alpha\beta^2)z + (\alpha\beta\gamma)^2$$

$$= z^3 - \left(\frac{1}{2} \right) z^2 + \alpha \beta \gamma (\beta + \alpha + \gamma) z + \alpha \beta \gamma$$

$$= z^3 + 2z^2 + 2z - 1 = 0$$

$$\alpha\beta + \beta\delta + \gamma\alpha = -2$$

$$\angle B(\alpha + \beta + \gamma) = 2$$

3

$$\alpha/\beta\gamma = 1.$$

$$\underline{3} \quad (a) \quad P(x) = (x-2)(x-3) Q(x) + ax+b \quad |$$

$$P(2)=4 \Rightarrow 4=2a+b$$

$$P(3)=9 \Rightarrow 9=3a+b$$

$$\text{Solve simultaneously} \Rightarrow \begin{aligned} a &= 5 \\ b &= -6 \end{aligned}$$

$$\therefore \text{remainder} = 5x-6 \quad |$$

$$(b) \quad (i) \quad \text{Let } P(x) = (x-\alpha)^n Q(x) \text{ where } Q(\alpha) \neq 0 \quad |$$

$$\text{then } P'(x) = (x-\alpha)^n \cdot Q'(x) + n(x-\alpha)^{n-1} Q(x) \quad |$$

$$= (x-\alpha)^{n-1} [(x-\alpha) Q'(x) + n Q(x)]$$

$$Q(\alpha) \neq 0 \quad \therefore P'(x) \text{ has } \alpha \text{ as a zero of multiplicity } n-1$$

$$(ii) \quad \text{Let } P(x) = ax^3 + bx^2 + d$$

$$\text{then } P'(x) = 3ax^2 + 2bx$$

$$P'(\alpha) = 0 \Rightarrow \alpha(3a\alpha + 2b) = 0$$

$$\alpha \neq 0 \text{ because } d \neq 0$$

$$\therefore \alpha = -\frac{2b}{3a} \quad |$$

$$\text{sub. in } P(x) \Rightarrow a\left(-\frac{2b}{3a}\right)^3 + b\left(-\frac{2b}{3a}\right)^2 + d = 0$$

$$\frac{-8b^3}{27a^2} + \frac{4b^3}{9a^2} + d = 0$$

$$\frac{-8b^3 + 12b^3}{27a^2} + d = 0$$

$$\frac{4b^3}{27a^2} + d = 0$$

$$\therefore 27a^2d + 4b^3 = 0 \quad |$$

$$\frac{4}{(a) (i)} \quad z^n = \cos n\theta + i \sin n\theta$$

$$z^{-n} = \cos(-n\theta) + i \sin(-n\theta) \\ = \cos n\theta - i \sin n\theta$$

(cos is an even function
sin is an odd function)

$$\therefore z^n + z^{-n} = \cos n\theta + i \sin n\theta + \cos n\theta - i \sin n\theta \\ = 2 \cos n\theta$$

$$(ii) \quad 2z^4 + 3z^3 + 5z^2 + 3z + 2 = 0$$

$z \neq 0 \quad \therefore$ divide both sides by z^2

$$\Rightarrow 2(z^2 + z^{-2}) + 3(z + z^{-1}) + 5 = 0$$

$$\text{i.e.} \quad 4 \cos 2\theta + 6 \cos \theta + 5 = 0$$

$$\text{i.e.} \quad 4(2\cos^2\theta - 1) + 6\cos\theta + 5 = 0$$

$$\text{i.e.} \quad 8\cos^2\theta + 6\cos\theta + 1 = 0$$

$$(4\cos\theta + 1)(2\cos\theta + 1) = 0$$

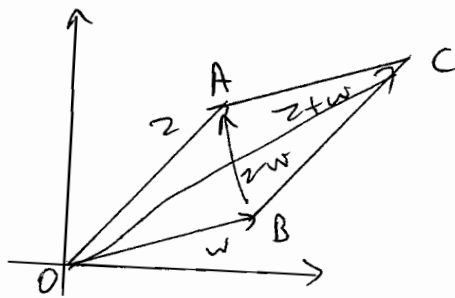
$$\therefore \cos\theta = -\frac{1}{4} \text{ or } \cos\theta = -\frac{1}{2}$$

$$\cos\theta = -\frac{1}{4} \Rightarrow \sin\theta = \pm \frac{\sqrt{15}}{4}$$

$$\cos\theta = -\frac{1}{2} \Rightarrow \sin\theta = \pm \frac{\sqrt{3}}{2}$$

$$\therefore z = -\frac{1}{4} \pm \frac{\sqrt{15}}{4}i \quad ; \quad -\frac{1}{2} \pm \frac{\sqrt{3}}{2}i$$

(b)



Let OA represent z and OB represent w

Then $z+w$ and $z-w$ are diagonals of parallelogram OACB

$$|z+w| = |z-w| \Rightarrow \text{diagonals are equal}$$

i.e. OACB is a rectangle

$$\text{i.e.} \quad \angle AOB = \frac{\pi}{2}$$

But $\angle AOB = \arg z - \arg w$

$$\text{i.e.} \quad \arg z - \arg w = \frac{\pi}{2}$$